Interpretability in Heterogeneous Treatment Effect Estimation Part I

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ACIC 2026 Short Course

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Link: `https://tinyurl.com/acic2026-hte`

Challenges with interpretability

In settings where researchers are dealing with multiple covariates, the CATE becomes challenging to interpret.

- Often in practice, estimates of $\hat{\tau}(X)$ are interpreted as individual-level treatment effects $\rightarrow$ Not correct! τ_i is fundamentally unidentifiable
- While HTE models output a prediction for the CATE, it can be useful to consider coarser representations of a conditional average treatment effect
- Examples of coarsened estimands:
  - Subgroup ATE
  $$\mathbb{E}\left[Y(1) - Y(0) \mid X \in \mathcal{C}(X)\right]$$
  - (Sorted) Group ATE (GATE)
  $$\mathbb{E}\left[Y(1) - Y(0) \mid Q_k(\tau(X)) \leq \hat{\tau}(X) \leq Q_{k+1}(\tau(X))\right]$$

We will focus primarily on the subgroup ATE.

How do we estimate subgroups?

Common approach: a priori specification of covariates, combined with linear regression.

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- Which covariates make sense to include?
- How do we think about interactions across these covariates?

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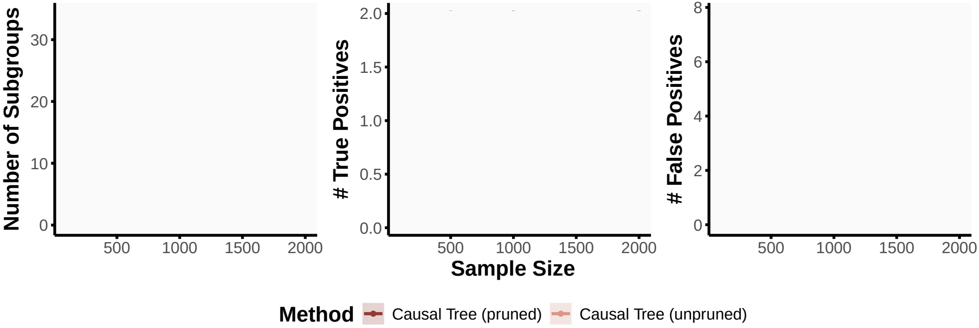

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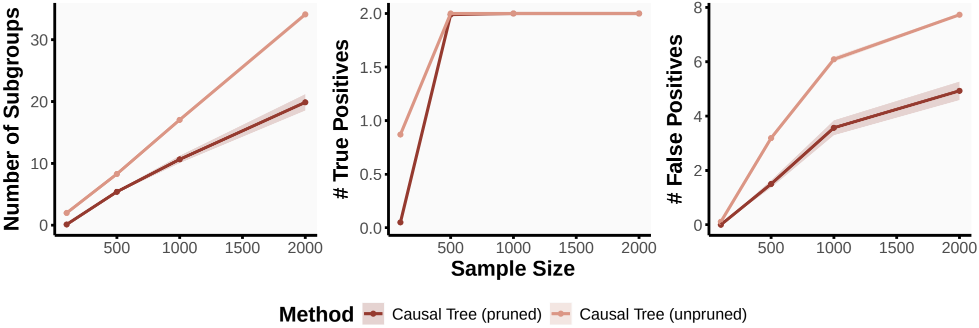

Goal: stable estimation of interpretable subgroups

Proposal: causal distillation trees to stably recover interpretable subgroups.

- Leverage the notion of knowledge distillation from machine learning to combine the benefits of black-box meta-learners with the interpretability of decision trees
- Unlike alternative two-stage approaches, we derive theoretical guarantees on the consistency of CDT in recovering the optimal subgroups
- To aid in practical usage, we introduce a novel model selection procedure to help researchers determine which meta-learner is most suitable for their setting

Causal Distillation Trees

Formalizing subgroups

We define a subgroup as $\mathcal{G}_g : \mathcal{X} \to \{0, 1\}$ where

$$\mathcal{G}_g(X) = \prod_{k=1}^{p} \mathbb{1}\left\{X^{(k)} \in R_k^{(g)}\right\}, \text{ and } R_k^{(g)} \subseteq \text{supp}(X^{(k)}).$$

- Example: $\mathcal{G}_g(X) = \mathbb{1}\{X^{\textsc{Age}} \geq 35\} \cdot \mathbb{1}\{X^{\text{College}} = 1\}$
- $\{\mathcal{G}_1(X_i), ..., \mathcal{G}_G(X_i)\}$ is a discrete partition.

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Causal distillation trees: An overview

Step 1: Estimate the first stage learner (the teacher model), to predict the conditional average treatment effect (i.e., $\hat{\tau}(X)$).

- Flexibility to use any black-box metalearner approach (e.g., X, T, R, DR-learners)
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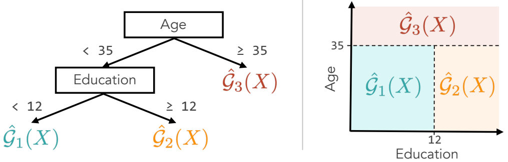
(A) Decision Tree **(B) Partition of $\mathcal{X}$**



(C) Subgroup Rules

$$\hat{\mathcal{G}}_1(X) = \mathbb{1}\{\text{Age} < 35\} \cdot \mathbb{1}\{\text{Education} < 12\}$$

$$\hat{\mathcal{G}}_2(X) = \mathbb{1}\{\text{Age} < 35\} \cdot \mathbb{1}\{\text{Education} \geq 12\}$$

$$\hat{\mathcal{G}}_3(X) = \mathbb{1}\{\text{Age} \geq 35\}$$

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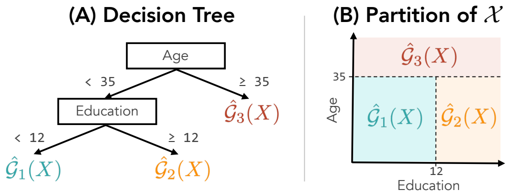

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Step 3: With $\hat{\mathcal{G}}(X)$, estimate the subgroup average treatment effect.

The role of sample splitting

To provide an honest estimate of the subgroup average treatment effect, we randomly split the data into a training and a hold-out estimation set (Athey and Imbens, 2016)

- The training set will be used to estimate the subgroups
  - To avoid overfitting to the teacher model, we estimate the $\hat{\tau}(X)$'s using out-of-sample procedures (e.g., out-of-bag samples in causal forest, or repeated cross fitting)
- Hold-out set reserved to estimate the subgroup average treatment effects.
- Allows us to employ standard approaches used in causal inference to estimate $\tau^{(g)}$

Recovering an optimal partitioning

Does CDT work? $\rightsquigarrow$ We want to know if CDT allows us to recover the optimal subgroups.

More formally, we define optimal subgroups as the set of subgroups $\mathcal{G}(X)$ such that

$$\{\mathcal{G}_g(X)\}_{g=1}^{G} = \underset{\{\mathcal{G}'_g(X)\}_{g\in\Omega}}{\operatorname{argmin}} \mathbb{E}\left[\left\{\tau(X_i) - \sum_g \mathcal{G}'_g(X_i)\cdot\tau^{(g)}\right\}^2\right],$$

where Ω denotes the space of partitions with fixed cardinality G of the covariate space.

- The optimal subgroups maximize treatment effect heterogeneity across groups
- Results can be generalized for alternative losses

CDT will consistently recover the optimal set of subgroups.

Assume consistency of the teacher model and separability. For all groups $g \in \{1, \ldots, G\}$:

$$\mathbb{E}\left(\left|\hat{\mathcal{G}}_g(X_i) - \mathcal{G}_g(X_i)\right|\right) \leq \frac{2r_g(C_\tau + M\delta)}{\delta} \underbrace{\left|\hat{s}_n^{(k)} - s^{(k)}\right|}_{\text{Gap between estimated and optimal split}} + \frac{2r_g}{\delta} \underbrace{\left\{\mathbb{E}_n\left[\ell(\hat{\tau}(X_i); X^{(k)}, s^{(k)})\right] - \mathbb{E}\left[\ell(\hat{\tau}(X_i); X^{(k)}, s^{(k)})\right]\right\}}_{\text{Gap between empirical and population-level loss}},$$

where r_g is the number of rules in the g^{th} subgroup, M is a finite positive constant, and δ corresponds to the constant in the separability condition.

- Immediate implication: $\hat{\mathcal{G}}_g(X_i) \xrightarrow{p} \mathcal{G}_g(X_i)$
- How quickly $\hat{\mathcal{G}}_g(X_i)$ converges to $\mathcal{G}_g(X_i)$ depends on how much reduction in noise we get from distilling

Improvements in stability from distillation

CDT allows researchers to more stably recover subgroups.

- The first-stage learner serves as a de-noising step for the observed data
- We construct a 'smooth' version of τ_i, which the decision tree is then estimated on

Recall the earlier toy example, with 2 subgroups:

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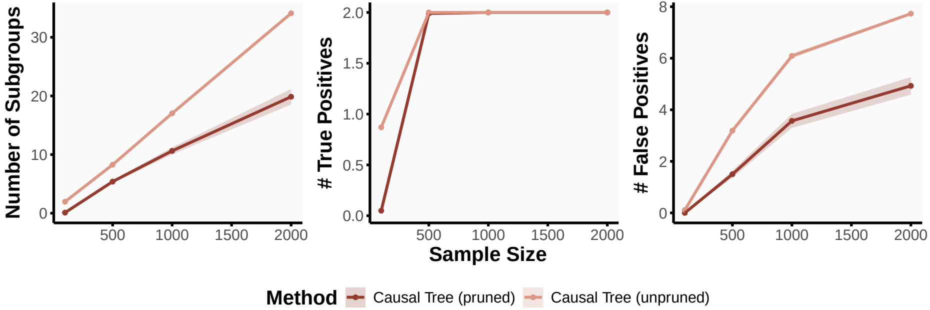

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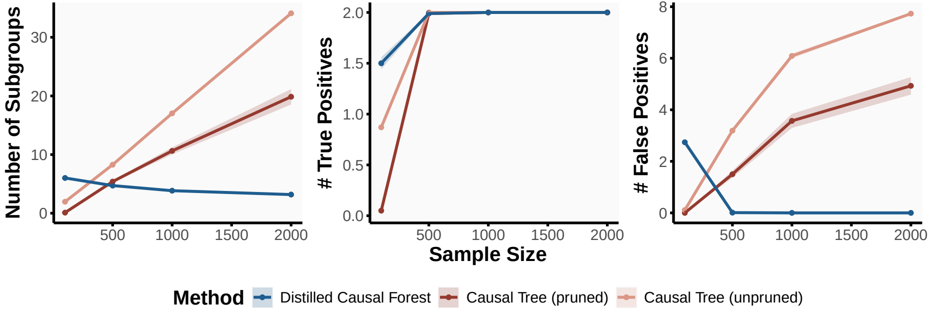
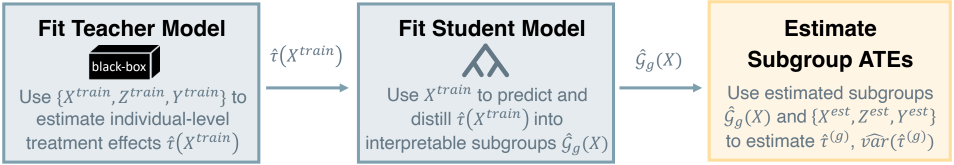


Toy Example

Summary

CDT offers a simple and flexible approach to estimating subgroups.



- Can accommodate federated learning, as well as settings in which we want to fit a student model with fewer covariates
- Distillation offsets instabilities in standard tree algorithms (like causal tree), while preserving interpretable output
- Sample splitting allows for straightforward inference on subgroup ATE

Choosing a teacher model

A stability-driven approach for teacher model selection

The benefits of distillation arise from the teacher model's ability to construct a meaningful representation of the conditional average treatment effect.

- In practice, there are different meta learners researchers must choose from to estimate the individual-level treatment effect.
- **How should researchers choose amongst a set of candidate teacher models?**

A stability-driven approach for teacher model selection

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We propose a model selection procedure tailored specifically for subgroup selection.

- Prioritize teacher models that result in stable subgroup reconstruction

  $\rightarrow$ Informally, this means being able to estimate similar subgroups, given perturbations in the underlying data generating process

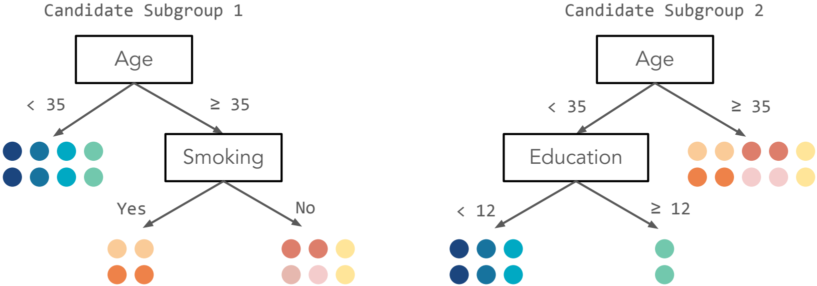
- Teacher models that are able to construct similar subgroups across perturbations are considered more appropriate models for the data (Yu, 2013; Yu and Kumbier, 2020; Rothenhäusler and Bühlmann, 2023)

Quantifying how similar subgroups are

Let $\hat{\mathcal{G}}^{(1)}$ and $\hat{\mathcal{G}}^{(2)}$ denote the two candidate sets of subgroups.

- Goal: we want to know how similar $\hat{\mathcal{G}}^{(1)}$ and $\hat{\mathcal{G}}^{(2)}$ are

Example:



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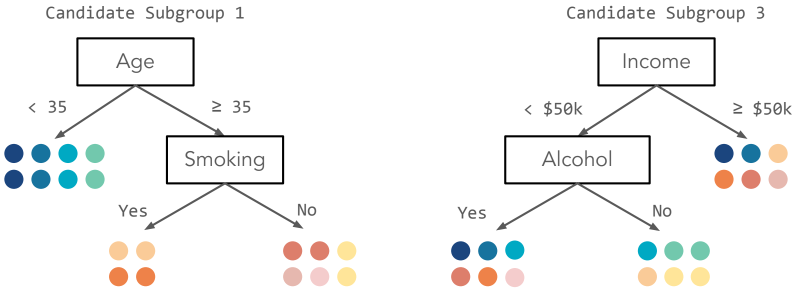


Subgroup Similarity Index (SSI)

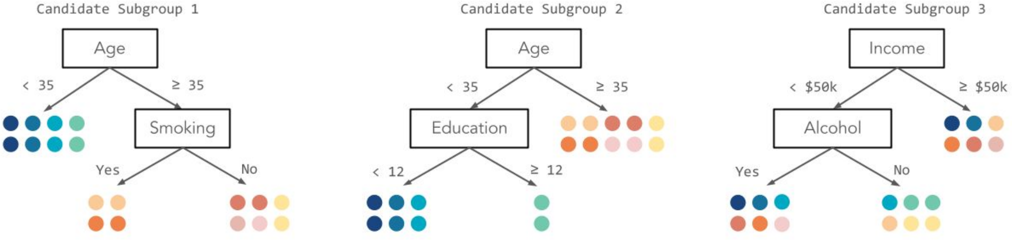
We propose a Jaccard Subgroup Similarity Index (SSI):

$$\text{SSI}(\hat{\mathcal{G}}^{(1)}, \hat{\mathcal{G}}^{(2)}) = \frac{1}{2G} \sum_{g \in \hat{\mathcal{G}}^{(1)} \cup \hat{\mathcal{G}}^{(2)}} \frac{\overbrace{N_{11}(g)}^{\#\text{ of sample pairs in same subgroup in both partitions}}}{\underbrace{N_{01}(g) + N_{10}(g)}_{\substack{\#\text{ of sample pairs in same subgroup in } \hat{\mathcal{G}}^{(j)} \\ \text{but different in } \hat{\mathcal{G}}^{(k)}}} + N_{11}(g)},$$

- SSI is be bounded between 0 and 1
- SSI $\rightarrow$ 1 implies the two sets of subgroups are similar.
  $\rightarrow$ Intuitively, will occur when $N_{11}(g)$ is much larger than $N_{01}(g) + N_{10}(g)$.
- In contrast, if two subgroups are highly dissimilar, then we would expect $N_{11}(g)$ to be much smaller than $N_{01}(g) + N_{10}(g)$ $\rightarrow$ low similarity index (SSI $\approx$ 0).

Quantifying how similar subgroups are

We can compute SSI for these three candidate subgroups:



- $\text{SSI}(\hat{\mathcal{G}}^{(1)}(X_i), \hat{\mathcal{G}}^{(2)}(X_i)) = 0.41$
- $\text{SSI}(\hat{\mathcal{G}}^{(1)}(X_i), \hat{\mathcal{G}}^{(3)}(X_i)) = 0.17$

Jaccard Illustration

Using SSI to select a teacher model

Consider a set of candidate teacher models $\{\hat{f}^1, \ldots, \hat{f}^L\}$.

- Generate $\hat{\tau}_i^d(l)$.
- For repeated bootstrap samples $b = 1, ..., 2B$:
  - Fit the second stage decision tree across the b and $b+1$-th samples ($\hat{f}^{(b)}_{student}$, $\hat{f}^{(b+1)}_{student}$) obtaining $\hat{\mathcal{G}}^{(b)}$ and $\hat{\mathcal{G}}^{(b+1)}$
  - Compute the Jaccard SSI: $\mathcal{J}_l^{(b)}$ =SSI($\hat{\mathcal{G}}^{(b)}, \hat{\mathcal{G}}^{(b+1)}$).

Averaging across all the bootstrap samples, choose the teacher model f^{l^*} that maximizes the Jaccard SSI:

$$l^* = \text{argmax}_{l=1,\ldots,L} \frac{1}{B} \sum_{b=1}^{B} \mathcal{J}_b^l$$

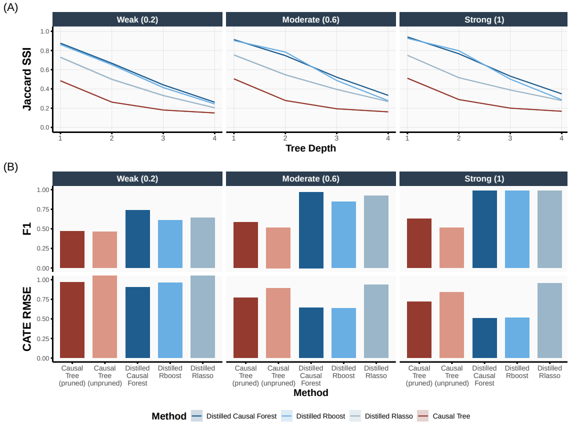
Illustration on simulated data

We compare SSI for (1) Causal Forest, (2) Rboost, and (3) Rlasso.

We benchmark to the stability of subgroups chosen by Causal Tree.

Key observations:

- Higher SSI corresponds to higher F1 score
  $\rightarrow$ Better accuracy in subgroup recovery
- Teacher models with higher SSI also have lower RMSE.



Empirical Application: AIDS Clinical Trial

Context

We consider the AIDS Clinical Trials Group Study 175 (ACTG 175) (Hammer et al., 1996).

- Randomized controlled trial where HIV-1 patients were assigned to either monotherapy (zidovudine alone) or combination therapy (ziduovudine and zalcitabine)
- Primary end point defined as:
  - A 50% or more decline in the CD4 cell count
  - The development of AIDS
  - Death

	Est. Average Treatment Effect (DiM)
Impact of Combination Therapy	0.13 (0.027)

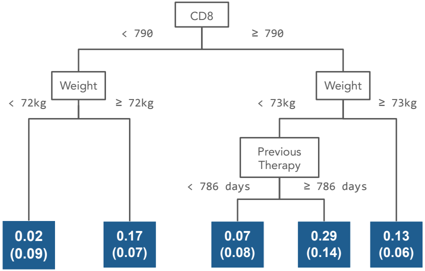
Treated Units: 524, Control Units: 532

Result: Combination therapy shown to slow the progression of HIV-1.
$\rightarrow$ Are there subgroups of the population that the treatment could be more effective for?

Subgroup analysis for AIDS Clinical Trials Group Study 175

Individuals with a combination of high CD8 counts (a healthy range is between 150-1000/mm3), starting weight, and little to no experience with anti-retroviral therapy.

- In general, those with higher CD8 counts strongly benefited regardless of weight or previous therapy.
- These benefits decrease especially as starting weight decreases.
- Treatments were later found to cause fat redistribution and fat loss that was harmful to patients



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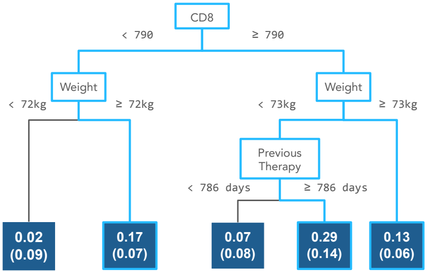
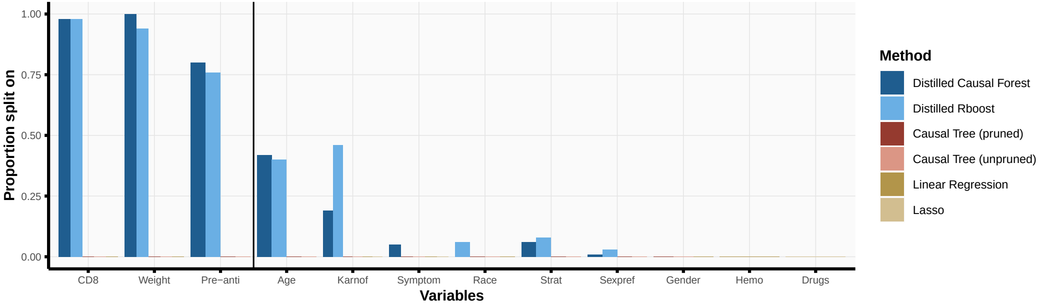
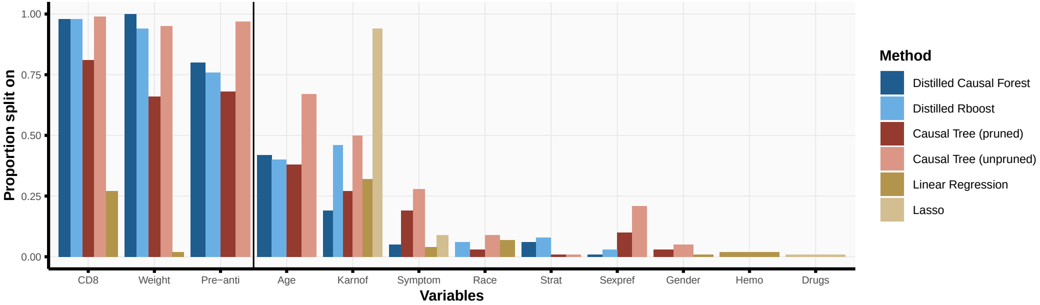


Stable and clinically relevant splits



- CD8 cell counts, weight, and previous experience with anti-retroviral therapy are all consistently selected across methods.
- However, competing approaches also tend to split on additional factors randomly.
- These do not all appear to be relevant!

Stable and clinically relevant splits



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- However, competing approaches also tend to split on additional factors randomly.
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Conclusion

We proposed causal distillation trees, which allows researchers to stably and consistently recover substantively meaningful subgroups.

- CDT allows researchers to integrate any metalearner of their choice and provides a simple, decision tree as an output
- The teacher model acts as a de-noising step which helps correct for instabilities in standard decision tree approaches
- Novel measure to perform teacher model selection

Next up...

- So far, we've discussed using a tree to distill information about $\hat{\tau}(X) \rightarrow$ What about alternative approaches?
- Variable importance and diagnostics?

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